

needs. A curriculum that has more of a practical orientation in addition to having theoretical perspective is required. There should be more stress on teaching the fundamentals and concepts. And it is necessary to arrange for guest lectures from industry veterans, so that the students could learn some of the best practices.

This is something which is also being recognised by the industry. Several companies have already taken proactive steps to increase the number of VLSI/embedded-trained engineers. Now we do see companies tying up with universities or setting up institutes to equip engineering graduates with the requisite skills.

elnfochips' university programme, *Impetus*, is designed to assist students by building their experience with embedded products, architecture and training materials. Students from engineering colleges are imparted training for four to six weeks.

TI India supports the academia through its University Programme. Termed as the *UniTI*, it stands for 'Universities and TI'. UniTI is the forum in which TI India interacts with universities and interested technical institutions in the role of a facilitator.

An ISA study on the status of semiconductor related research in Indian universities reveals that:

1. The role of government funding is 85-90 per cent of the funding available to university lab research
2. The top seven institutes account for 70 to 75 per cent of the total

A significant trend is the move towards small form-factor designs which can be upgraded with minimal application efforts.

—Nirav Shah, director of marketing, elnfochips



research being undertaken in the academic institutes of the country. These institutes have the faculty and dedicated VLSI labs to support high-quality semiconductor-related research

3. These institutes account for around 61 per cent of research papers published in the field of semiconductors

The study recommends:

1. A public-private partnership (PPP) model involving the government, the industry and the academia to encourage applied semiconductor research
2. Setting up of technology business incubators (TBI). TBIs provide a robust platform for active industry-academia interactions, and help convert potential research ideas into their commercial success
3. Steps to enhance research infrastructure by developing dedicated research centres, enhancing the standard of semiconductor laboratories to match with industry standards and setting up semiconductor-related research centres in special economic zones

4. Performance-based incentives to faculty and research scholars, facilitating involvement of industry executives in academics and creating awareness about job avenues among PhD holders

5. Participation of the industry beyond just providing tools

Road to success

The growth in the next four years for the embedded service industry will be higher in companies offering focused solutions on industry segments like automotive, industrial, portable/wireless and consumer electronics.

The future of embedded software looks bright given the exceeding pressure on the US companies for time-to-market and cost. Indian companies are investing in infrastructure and building domain expertise. IP creation will lead to royalty-based models as well as high-end expertise in specific domains in the embedded software value chain.

Having a local manufacturing ecosystem will also boost this industry with end-to-end product development and rollout happening from India. Companies that have been providing embedded design services will move up the value chain by building reusable components and undertaking complete system development, i.e., all levels of software development. There will be a strong focus on increasing margins and adopting business models which are not linearly dependent on the manpower. ●

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New applications would be e-passports, smartcard-based rural banking and micro-finance, live-stock identification and identification-based vehicle security.

—Dr N.S. Murthy, director, new business initiatives, NXP Semiconductors

